TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

PULCHOWK CAMPUS

STUDY OF EMPLOYEE SCHEDULING PROBLEM IN ARPAN BANQUTTE BANESHOWR, KATHMANDU

Submitted By:

Anusha Acharya (074BME607)

Prajwal Koirala (074BME627)

A PROJECT REPORT

SUBMITTED TO THE DEPARTMENT OF MECHANICAL AND AEROSPACE ENGINEERING IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR IN MECHANICAL ENGINEERING

DEPARTMENT OF MECHANICAL AND AEROSPACE ENGINEERING

LALITPUR, NEPAL

DECEMBER, 2021

ABSTRACT

Many professions require a certain number of employees to work at a certain time. Optimal scheduling of employees can have tremendous benefits including increased productivity, employee satisfaction, minimized payroll costs, and many more. This report presents a solution for scheduling employees by formulating it as an optimization problem. The first and foremost step is to mathematically model the problem, with a clear definition of decision variable, objective function, and constraints. The readily available solver like Simplex in MS. Excel was used to find the solution of an example problem.

ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to our late teacher Asst. Prof. Mr. Looza Shrestha for introducing and guiding us to the Operational Research course. Without his valuable guidance, this project would not have been completed.

Also, we would like to express our deepest gratitude to our faculty teacher Asst. Prof. Sanjay Neupane and Asst. Prof. Ayush Bhattarai for sharing their valuable time, knowledge, and guidance in the moment of crisis and giving us the opportunity to continue with this project.

Team Members

Anusha Acharya (074BME607)

Prajwal Koirala (074BME627)

TABLE OF CONTENTS

ABSTRACT ii

ACKNOWLEDGEMENTS iii

TABLE OF CONTENTS iv

LIST OF FIGURES v

LIST OF TABLES vi

CHAPTER ONE: INTRODUCTION 1

1.1. Background 1

1.2. Problem Statement 2

1.3. Objectives 2

CHAPTER TWO: METHODOLOGY 3

2.1. Problem Defination 4

2.2. Linear Programming Problem Formulation 5

2.3. Excel Formulation 6

2.4. Solution 7

2.5. Staff Assignment Problem: 7

CHAPTER THREE: CONCLUSION 9

References 10

LIST OF FIGURES

Figure 2.1 Methodolgy Chart 4
Figure 2.2 Problem Formulation in Excel 6
Figure 2.3 Staff assignment problem formulation in Excel 8

LIST OF TABLES

Table 2.1 Staffs Requirement 4
Table 2.2 Employee work schedules 4
Table 2.3 Optimal values of decision variables 7

CHAPTER ONE: INTRODUCTION

1.1. Background

Employee scheduling problems arise in various fields such as health care, transportation, call centers, hotel and airline reservation, manufacturing, protection services, and others where employees are required to cover different shifts for the fulfillment of organizational objectives.

The scheduling of employee problems can come in different forms like assigning employees to certain shifts as in the case of nurse staff or to design shifts such that there is always a presence of a certain employee, or to design shifts to cover some given tasks. (Kletzander and Musliu, 2020). A simple form of scheduling problem includes the scheduler needing to give a certain day off to the employers who work standard shifts with different starting time, often assuming the required number of employers are on duty throughout the day. Brownell and Lowerre (1976) presents a variation of this type of problem. Shift scheduling like in telephone operators is much more complex which involves part-time working as well as full-time working employees. Their main objective is often to determine approximately the desired number of employees to be on duty by minimizing the "overage". Morris and Showalter (1983) presents a general scheduling problem considering both complex days off and shift scheduling along with considering employee tours.

Assigning employees to shifts is not an easy task. Legal obligations like weekly holidays, extra paid for overtime works, etc. as mentioned by the company or the government has to be fulfilled. Even more so, sometimes employees might specify their shift for a special ocassion, therefore, the fairness for employees and their satisfaction are also to be considered. Often scheduling problems have certain definite objectives like minimizing wages or maximizing productivity but are faced with a wide range of constraints. In a typical staffing problem, the constraint might include minimum and maximum hours to be worked during the week.

This report focuses on solving staff assignment problems of Arpan Banquet located in Baneshwor, Nepal by considering a smaller number of decision variables. However, the problems arising in other industries are much more involved. Glover and McMillan (1986)

presents the integration of MS and AI to solve the general employee scheduling problem with increased complexity.

1.2. Problem Statement

The lack of proper scheduling may lead to unnecessary overtime working of employees, increased payroll cost, decreased productivity, overlapped works, employees dissatisfaction and many more. For instance the decision about the number of nurses to be employed at a certain time has a great effect on hospital costs as they comprise a major portion of hospital staff with significant wages. A similar case example is considered in our project wherein a Banquet needs to decide on the number of waiters to recruit and along with their time table so as to minimize the total payroll cost.

1.3. Objectives

The main objective of this project is:

- To formulate employee scheduling problems as an optimization problem and find the solution using available solvers (in MS. Excel).

CHAPTER TWO: METHODOLOGY

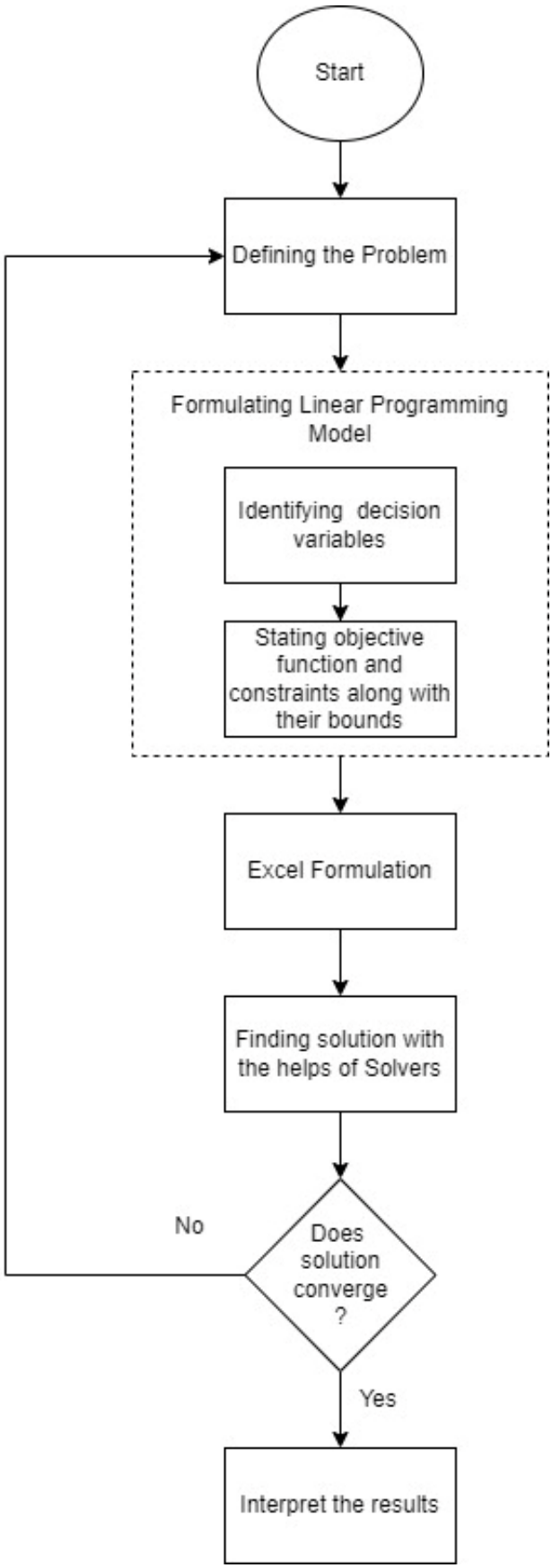

Figure 2.1 Methodolgy Chart

2.1. Problem Defination

Arpan Banquet, Baneshwor needs a certain number of waiters each day of the week as given in the table.

Every staff gets paid Rs.500/day and those who work on Saturday get paid Rs.200 extra.

Each staff must be on a schedule that allows him or her to take two days off in a row. To reduce total payroll costs every week, the Banquet must decide how many waiters to recruit and on what timetable they should work. Employees are hired on a contract basis rather than part-time for security reasons.

Table 2.1 Staffs Requirement

Day	Number of staffs needed
Sunday	12
Monday	14
Tuesday	14
Wednesday	10
Thursday	10
Friday	14
Saturday	17

As indicated in the table below, we can define seven different work schedules for an employee:

Table 2.2 Employee work schedules

Schedule number	Working Days	Days off	No. of staffs
S1	Tuesday to Saturday	Sunday, Monday	x_1

S2	Wednesday to Sunday	Monday, Tuesday	x_2
S3	Thursday to Monday	Tuesday, Wednesday	x_3
S4	Friday to Tuesday	Wednesday, Thursday	x_4
S5	Saturday to Wednesday	Thursday, Friday	x_5
S6	Sunday to Thursday	Friday, Saturday	x_6
S7	Monday to Friday	Saturday, Sunday	x_7

2.2. Linear Programming Problem Formulation

Decision Variables:

Number of employees per each schedule are x_1, x_2, x_3, x_4, x_5, x_6 and x_7 . These are the 7 decision variables in this staffing problem each corresponding to different schedules as enlisted in the table.

Constraints:

The number of employees can be neither negative nor fractions; clearly, this is an integer programming problem. So,

$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0$

$x_1, x_2, x_3, x_4, x_5, x_6, x_7$ are integers

Each day, Arpan Banquet requires a sufficient number of personnel to operate the rides.

For example, to ensure enough waiters on Sunday, the following constraint is imposed.

$x_2 + x_3 + x_4 + x_5 + x_6 \geq 12$

Similarly, for other days in the week:

Monday: $x_3 + x_4 + x_5 + x_6 + x_7 \geq 14$

Tuesday: $x_4 + x_5 + x_6 + x_7 + x_1 \geq 14$

Wednesday: $x_5 + x_6 + x_7 + x_1 + x_2 \geq 10$

Thursday: $x_6 + x_7 + x_1 + x_2 + x_3 \geq 10$

Friday: $x_7 + x_1 + x_2 + x_3 + x_4 \geq 14$

Saturday: $x_1 + x_2 + x_3 + x_4 + x_5 \geq 17$

Objective Function:

The waiters who follow the working schedule S6 and S7 do not work on Saturday. So, they earn Rs. 2500 per week. Similarly, those who work according to schedule S1, S2, S3, S4 and S5 earn an extra bonus of Rs. 200. So, they earn Rs. 2700 per week. Therefore, the objective function of this problem is:

Min $J = 2700 * (x_1 + x_2 + x_3 + x_4 + x_5) + 2500 * (x_6 + x_7)$

2.3. Excel Formulation

	A	B	C	D	E	F	G	H	I	J
1										
2	Schedule	Staff count	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	
3	S1	5	0	0	1	1	1	1	1	
4	S2	0	1	0	0	1	1	1	1	
5	S3	5	1	1	0	0	1	1	1	
6	S4	2	1	1	1	0	0	1	1	
7	S5	5	1	1	1	1	0	0	1	
8	S6	0	1	1	1	1	1	0	0	
9	S7	2	0	1	1	1	1	1	0	
10		Total	12	14	14	12	12	14	17	
11		Required	12	14	14	10	10	14	17	
12										
13		Total waiters required	19							
14		Pay/waiter/week	2500							
15		Saturday Bonus	200							
16		Total Pay per Week	50900							
17										

Figure 2.2 Problem Formulation in Excel

The staffing problem is formulated in excel as shown in the figure. The cells $B$3:$B$9 are the decision variables and the cell $C$16 for total pay per week is the objective function.

Excel formula for C10 = SUMPRODUCT(C3:C9,$B$3:$B$9)

Excel formula for B13 = SUM(B3:B9)

Excel formula for C16 = SUM(C14:C15)*SUM(B3:B7)+ SUM(B8:B9)*C14

The solver setting for this optimization problem:

Minimize: $C$16

Decision variables: $B$3:$B$9

Constraints: $C$10:$I$10 ≥ $C$11:$I$11

$B$3:$B$9 = Integer

$B$3:$B$9 ≥ 0

2.4. Solution

Upon solving the above problem, following optimal values of the decision variables are obtained:

Table 2.3 Optimal values of decision variables

Schedule number	Decision Variable	Optimal Value
S1	x_1	5
S2	x_2	0
S3	x_3	5
S4	x_4	2
S5	x_5	5
S6	x_6	0
S7	x_7	2

2.5. Staff Assignment Problem:

Arpan Banquet currently has 25 staff working as waiters. Each of the employees have individual ratings based on their track record. They are asked to fill a form prioritising the available work schedule S1, S3, S4, S5 and S7. Based on the employee preferences the employees are assigned to different schedules. So, the sum of employee preferences weighted by their track record is to be maximized while the employee requirement per

schedule is to be met. Unfortunately, six of the employees will be made redundant to restrict the total staff count to 19.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1														
2														
3			Staff's Priority					Staff Assignment						
4	Employee	Rating (on 10)	S1	S3	S4	S5	S7	S1	S3	S4	S5	S7	Total	Sum
5	E1	8	4	5	3	2	1	0	1	0	0	0	5	1
6	E2	1	1	3	2	4	5	0	0	0	0	0	0	0
7	E3	5	5	4	3	2	1	1	0	0	0	0	5	1
8	E4	4	3	5	4	1	2	0	0	1	0	0	4	1
9	E5	8	5	2	4	3	1	1	0	0	0	0	5	1
10	E6	1	2	1	3	5	4	0	0	0	0	0	0	0
11	E7	3	1	5	4	3	2	0	0	0	1	0	3	1
12	E8	6	4	5	3	2	1	0	1	0	0	0	5	1
13	E9	1	5	2	4	3	1	0	0	0	0	0	0	0
14	E10	4	1	3	2	4	5	0	0	0	1	0	4	1
15	E11	7	4	3	2	5	1	0	0	0	1	0	5	1
16	E12	3	3	5	4	1	2	0	0	1	0	0	4	1
17	E13	4	2	1	3	5	4	0	0	0	1	0	5	1
18	E14	9	2	4	3	1	5	0	0	0	0	1	5	1
19	E15	2	1	3	2	4	5	0	0	0	0	0	0	0
20	E16	7	1	5	4	3	2	0	1	0	0	0	5	1
21	E17	9	3	5	4	1	2	0	1	0	0	0	5	1
22	E18	8	5	2	4	3	1	1	0	0	0	0	5	1
23	E19	9	1	4	2	3	5	0	0	0	0	1	5	1
24	E20	7	4	3	2	1	5	1	0	0	0	0	4	1
25	E21	5	5	3	1	2	4	1	0	0	0	0	5	1
26	E22	8	3	2	1	5	4	0	0	0	1	0	5	1
27	E23	4	3	5	4	2	1	0	1	0	0	0	5	1
28	E24	2	1	4	2	3	5	0	0	0	0	0	0	0
29	E25	1	2	4	3	1	5	0	0	0	0	0	0	0
30														
31														
32							Total	5	5	2	5	2	566	19
33							Required	5	5	2	5	2		
34														

Figure 2.3 Staff assignment problem formulation in Excel

Excel formulation of this staff assignment problem is as shown in the figure.

Solver settings:

Minimize: $M$32

Decision variables: $H$5:$L$29

Constraints: $H$32:$L$32 = $H$33:$L$33

$H$5:$L$29 ≤ 1

$H$5:$L$29 ≥ 0

$N$5:$N$29 ≤ 1

$H$5:$L$29 = integer

Upon solving the above problem, assignment of each waiter to the schedules S1, S3, S4, S5, S7 is completed. Employees E2, E6, E9, E15, E24 and E25 are not assigned to any of the schedules. Remaining others are assigned based on their preferences weighted by their rating such that the staff requirement per schedule is fulfilled.

CHAPTER THREE: CONCLUSION

Staff scheduling problem followed by assignment was performed as per the requirement of Arpan Banquet, Baneshwor. The linear programming problem was formulated and modeled in a spreadsheet and solved using excel solver. The optimal number of waiters required per month was determined to be 19. Each of these employees were assigned to the available working schedules based on their preferences and track record.

REFERENCES

Brownell, W. S., & Lowerre, J. M. (1976). Scheduling of work forces required in continuous operations under alternative labor policies. *Management Science, 22*, 597--605.

Glover, F., & McMillan, C. (1986). The general employee scheduling problem. An integration of MS and AI. *Computers & operations research, 13*, 563-573.

Kletzander, L., & Musliu, N. (2020). Solving the general employee scheduling problem. *Computers & Operations Research, 113*, 104794.

Morris, J. G., & Showalter, M. J. (1983). Simple approaches to shift, days-off and tour scheduling problems. *Management Science, 29*, 942-950.